Computing Convolution:

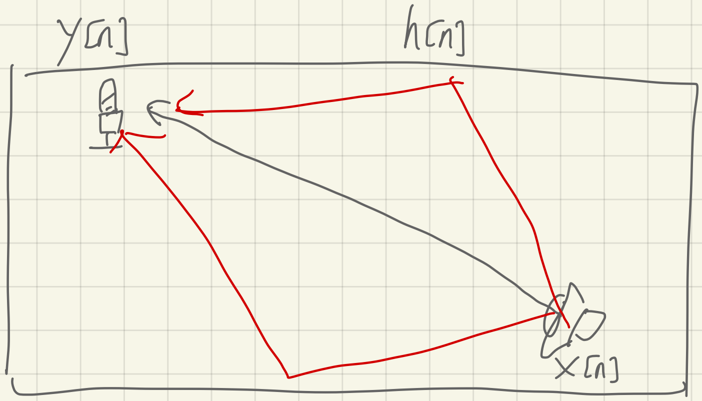
3. **Matrix-vector multiply** (for finite duration signals)

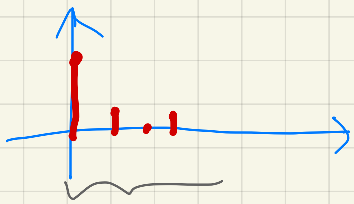
$$y[n] = \sum_k x[k]\, h[n-k] = \sum_k h[k]\, x[n-k]$$

Example: $M = 3$; $x[n]$ zero for $n < 0$ and $n > 7$.

Toeplitz matrix.

$$\begin{bmatrix} x[0] & 0 & 0 & 0 \\ x[1] & x[0] & 0 & 0 \\ x[2] & x[1] & x[0] & 0 \\ x[3] & x[2] & x[1] & x[0] \\ x[4] & x[3] & x[2] & x[1] \\ x[5] & x[4] & x[3] & x[2] \\ x[6] & x[5] & x[4] & x[3] \\ x[7] & x[6] & x[5] & x[4] \\ 0 & x[7] & x[6] & x[5] \\ 0 & 0 & x[7] & x[6] \\ 0 & 0 & 0 & x[7] \end{bmatrix} \begin{bmatrix} h[0] \\ h[1] \\ h[2] \\ h[3] \end{bmatrix} = \begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \\ y[4] \\ y[5] \\ y[6] \\ y[7] \\ y[8] \\ y[9] \\ y[10] \end{bmatrix}$$





Q: given $x[n]$ & $y[n]$

how do we find $h[n]$?

$$y[n] = h[n] * x[n]$$

We also know the length of $h[n]$.

$$\vec{y} \cong \underbrace{T_{\vec{x}}}_{\text{Toeplitz matrix}} \vec{h}$$

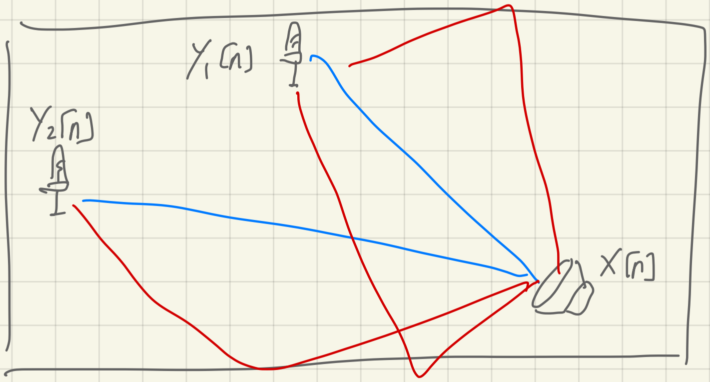
$$\min_{\vec{h}} \| \vec{y} - T_{\vec{x}} \vec{h} \|_2^2$$

least squares

Let $A = T_{\vec{x}}$

$$\min_{h} \| y - Ax \|_2^2$$

$$\hat{x} = A^{\dagger} y = (A^H A)^{-1} A^H y$$



$y_1[n] = h_1[n] * x[n]$

$y_2[n] = h_2[n] * x[n]$

$x[n]$: unknown

$h_1[n], h_2[n]$: only length is known

$y_1[n], y_2[n]$: observed.

$$\begin{aligned}
y_1[n] * h_2[n] &= (h_1[n] * x[n]) * h_2[n] \\
&= (x[n] * h_1[n]) * h_2[n] \qquad \because \text{commutativity} \\
&= x[n] * (h_1[n] * h_2[n]) \\
&= x[n] * (h_2[n] * h_1[n]) \\
&= (x[n] * h_2[n]) * h_1[n] \\
&= y_2[n] * h_1[n]
\end{aligned}$$

$T_{y_1} \vec{h_2} = T_{y_2} \vec{h_1}$ matrix form

$$\underbrace{\begin{bmatrix} T_{y_2} & -T_{y_1} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} \vec{h_1} \\ \vec{h_2} \end{bmatrix}}_{\vec{x}} = \vec{0}$$

$\vec{x}$ is a null vector of A

$B \in \mathbb{C}^{n \times n}$ $\quad \vec{x} \in \mathbb{C}^n$

If there exists $\lambda \in \mathbb{C}$ such that $B\vec{x} = \lambda \vec{x}$

then $\vec{x}$ is an eigenvector of B with eigenvalue λ.

$A\vec{x} = \vec{0}$ $\quad$ A: rectangular

$\underbrace{A^H A}_{\text{always square}} \vec{x} = \vec{0}$ $\quad$ "eig" in Matlab

"eigs"

power method.

$y[n] = h[n] * x[n]$ $\quad$ $\vec{h} \in \mathbb{C}^L$ $\quad$ $\vec{x} \in \mathbb{C}^M$

$\boxed{\vec{y} = T_h \vec{x}}$ $\quad$ $\Rightarrow \vec{y} \in \mathbb{C}^{M+L-1}$

$$y[n] = \sum_{m=-\infty}^{\infty} h[m]\, x[n-m]$$

T_h is a linear mapping from $\mathbb{C}^M$ to $\mathbb{C}^{M+L-1}$

T_h is a linear operator

$x[m,n]$ 2D signal. $\quad$ $y[m,n]$

$$h[m,n] * x[m,n] = \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} h[k,l]\, x[m-k, n-l]$$

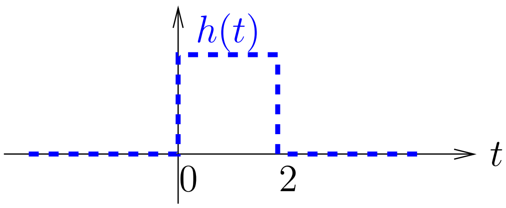
"conv2"

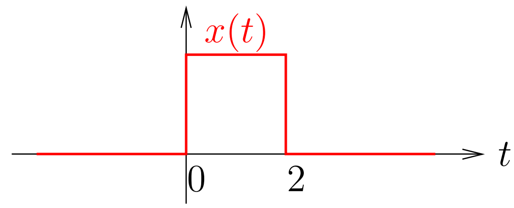
$$y_2[m,n] * \underbrace{h_1[m,n]}_{\vec{h}} = y_1[m,n] * h_2[m,n]$$

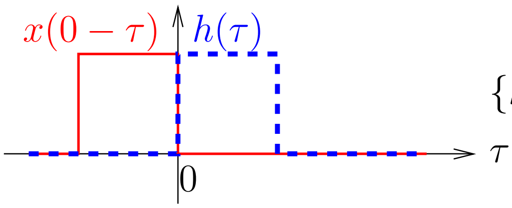
$$\begin{bmatrix} T_{y_2} & -T_{y_1} \end{bmatrix} \begin{bmatrix} \vec{h}_1 \\ \vec{h}_2 \end{bmatrix} = 0$$

CT Convolution graphically:

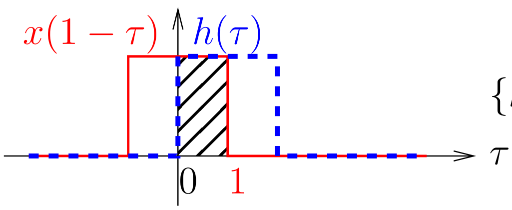
Graphically, convolution can be understood as “flip and slide”:



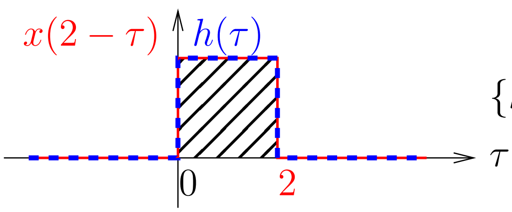




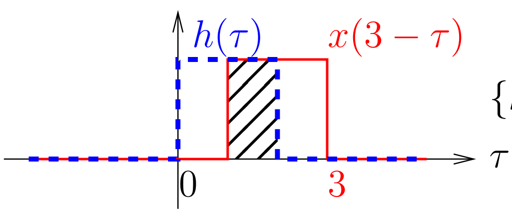
$$\{h(t) * x(t)\}_{t=0} = \int_{-\infty}^{\infty} h(\tau)x(0-\tau)d\tau$$



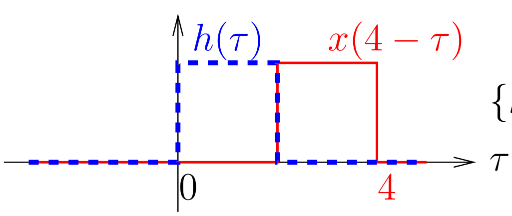
$$\{h(t) * x(t)\}_{t=1} = \int_{-\infty}^{\infty} h(\tau)x(1-\tau)d\tau$$



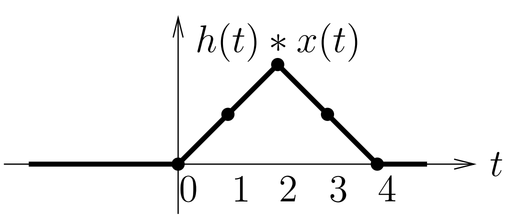
$$\{h(t) * x(t)\}_{t=2} = \int_{-\infty}^{\infty} h(\tau)x(2-\tau)d\tau$$



$$\{h(t) * x(t)\}_{t=3} = \int_{-\infty}^{\infty} h(\tau)x(3-\tau)d\tau$$



$$\{h(t) * x(t)\}_{t=4} = \int_{-\infty}^{\infty} h(\tau)x(4-\tau)d\tau$$



LTI systems — convolution:

Some important properties of convolution:

1. Commutative: $x * h = h * x$
   [block diagrams]

2. Associative: $(x * h) * w = x * (h * w)$
   [block diagrams]

3. Distributive: $x * (h + w) = x * w + x * h$
   [block diagrams]

4. Identity: $h * \delta = h$
   [block diagrams]

$a, b \in \mathbb{R}$

$ab = ba$

$(ab)c = a(bc)$

$a(b+c) = ab + ac$

$a \cdot 1 = a$

These apply in both discrete-time and continuous-time cases.

LTI systems — causality:

- An LTI system is *causal* iff $h[n] = 0$ for all $n < 0$
- Discrete-time case:

  $\Leftarrow$ (Assume $h[n] = 0$ for all $n < 0$ and show causal.)

  For an LTI system, we know

  $$\begin{aligned} y[m] &= \textstyle\sum_{n=-\infty}^{\infty} h[n]x[m-n] \\ &= \textstyle\sum_{n=0}^{\infty} h[n]x[m-n] \text{ (since } h[n]|_{n<0} = 0\text{),} \end{aligned}$$

  so $y[m]$ depends only on $x[m], x[m-1], x[m-2], \ldots$.

  Thus, system is causal.

  $\Rightarrow$ (Assume causal and show $h[n] = 0$ for all $n < 0$, or assume $h[n] \neq 0$ for some $n < 0$ and show noncausal.)

  For an LTI system,

  $$y[m] = \textstyle\sum_{n=-\infty}^{\infty} h[n]x[m-n].$$

  Assuming $h[-d] \neq 0$ for some $d > 0$, can see that $y[m]$ depends on the future value $x[m+d]$.

  Thus, the system is noncausal.

- Continuous-time case: same idea...

LTI systems — stability:

- An LTI system is (BIBO) *stable* iff its impulse response is *absolutely summable* ($\sum_n |h[n]| < \infty$ or $\int |h(t)| dt < \infty$).

  Can we prove this?

- Discrete-time case:

  $\Leftarrow$ (Assume $\sum_n |h[n]| < \infty$ and show system is stable.)
  For an LTI system, know

$$\begin{aligned} |y[n]| &= \Big|\sum_m x[m]h[n-m]\Big| \\ &\leq \sum_m |x[m]h[n-m]| \\ &= \sum_m |x[m]|\,|h[n-m]|. \end{aligned}$$

  and if input is bounded (i.e., $|x[m]| < M_x$ for all m),

$$< M_x \underbrace{\sum_m |h[n-m]|}_{<\infty},$$

  so the output is also bounded.

  $\Rightarrow$ (Assume system is stable and show $\sum_n |h[n]| < \infty$, or assume $\sum_n |h[n]| = \infty$ and show system is unstable.)
  Consider the bounded input $x[m] = \operatorname{sgn}(h[-m])$:

$$\begin{aligned} y[n] &= \sum_m \operatorname{sgn}(h[-m])h[n-m] \\ y[0] &= \sum_m |h[-m]| = \infty. \end{aligned}$$

  Since the output is unbounded, the system is unstable.

- Continuous-time case: same idea. . .

An algebra for LTI systems?:

- We've seen that an LTI system's output can be written as the convolution of the input and the impulse response:

$$\mathcal{H}\{\{x[n]\}\} = \sum_{m=-\infty}^{\infty} x[m]\{h[n-m]\}$$

$$\mathcal{H}_{\mathsf{c}}\{\{x(t)\}\} = \int_{-\infty}^{\infty} x(\tau)\{h(t-\tau)\}d\tau.$$

- But integrals and infinite-sums are a pain! Is there an "*algebra*" for signals and systems? (Algebra involves only multiplications & additions.)

  Yes! Through the "frequency domain"...

- We first review the details for continuous-time (CT) systems, and later proceed to discrete-time (DT) systems.

- As we shall see, there are many forms of the "*frequency domain*", including Fourier series, the CTFT, the Laplace transform, the DTFT, the DFT, and the Z-transform. Understanding the relationships among these different "frequency" representations is one of the primary objectives of this course.